

# Dev Trivedi

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## Overview

Computer Science graduate student experienced in backend engineering and real-time AI-powered web systems. Proficient in Python, Java, REST APIs, SQL databases, and deploying scalable applications integrating computer vision and machine learning models.

## Education

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|-----------------------------------------------------------------------------------------|----------------------|
| <b>California State University, Long Beach</b> – Masters of Science in Computer Science | Aug 2025 - May 2027  |
| <b>Gujarat Technological University</b> – B.Tech in Computer Science Engineering        | Sept 2021 - May 2025 |

## Skills

**Languages:** Python, Java, C++, C, JavaScript, SQL, TypeScript  
**Backend and Databases:** Django, Flask, FastAPI, Spring Boot, Node.js, MySQL, PostgreSQL, MongoDB  
**Frontend:** React, Next.js, Tailwind CSS, Material UI  
**Computer Vision & ML:** OpenCV, YOLOv8, NumPy, Model Evaluation, Real-Time Inference  
**Tools & Platforms:** Git, Docker, AWS, GCP, Vercel, Railway, Cloudinary, GitHub Actions  
**Core Concepts:** Data Structures & Algorithms, Software Design, OOP, Multithreading, Distributed Systems, AI/Machine Learning Integration, Model Evaluation, Automated Testing, CI/CD

## Experience

- Computer Vision & Autonomous Systems Engineer (BAD)** – California State University, Long Beach    Feb 2026 – Present
- Developed OpenCV and YOLOv8 pipelines for object detection and autonomous mission execution on quadcopter for UASC
  - Configured Pixhawk and Mission Planner for autonomous waypoint navigation and recalibrated RC, servos, and LiDAR across quadcopter and thermal UAV platforms.
  - Collaborated with a multidisciplinary team to develop and field-test autonomous UAV systems for competition readiness.
- Software Development Intern** – Zeitech IT Services Pvt. Ltd.    Jan 2025 – April 2025
- Built and maintained backend microservices using REST APIs, improving scalability and reducing response times by 30%.
  - Built and maintained Python-based internal APIs and scripts for data processing, API testing, and workflow automation, improving developer efficiency.
  - Designed multithreaded endpoints, sustaining 200+ concurrent requests at <250ms latency in load tests.
- Full Stack Developer** – Baxa Agency Pvt. Ltd.    Aug 2024 – Dec 2024
- Developed a Spring Boot + React real-estate platform, improving API latency for 450+ listings through query optimization.
  - Built a marketing site as a high-performance React/Node web app, boosting page-load speed 40% and improving SEO rank.
  - Implemented structured logging, health checks, and unit testing suite around critical APIs, improving uptime by 15%.

## Projects

- Real-Time Vehicle Detection & Traffic Analysis System** | Python, OpenCV, YOLOv8, Flask, NumPy
- Developed real-time vehicle detection system using OpenCV and YOLOv8 to detect and classify cars, trucks, and motorcycles from traffic video streams.
  - Implemented object tracking and line-crossing logic to count vehicles and estimate traffic density.
  - Built a Flask-based web server to stream processed video frames and expose detection results via REST endpoints.
- Golden Nest – Real Estate Platform** | Java, Spring Boot, MySQL, React, Git/Github, Railway | [GoldenNest.com](https://GoldenNest.com)
- Built a full-stack real estate platform using Spring Boot REST APIs and React for property listing and search.
  - Optimized MySQL queries with indexing and pagination, reducing search latency by ~30%.
  - Implemented JWT-based authentication and role-based authorization to secure endpoints and protect workflows.
  - Deployed via Docker and Railway, ensuring reliable builds, clean architecture, and seamless frontend–backend integration.
- Creative Baxa – Agency Website** | React, TypeScript, Tailwind CSS, Node.js, Vercel | [CreativeBaxa.com](https://CreativeBaxa.com)
- Built and deployed production-grade website using React + TypeScript with modular, reusable component architecture.
  - Implemented responsive UI using Tailwind CSS, ensuring cross-device compatibility and optimized accessibility scores.
  - Integrated backend services using Node.js APIs for dynamic content handling and structured form submissions.